

P.P (2023) ODE

Q101:-

$$\frac{dy}{dx} = 1+x+y^2+xy^2, \quad y(1)=0$$

$\downarrow$ $\downarrow$
 x y

$$\frac{dy}{dx} = (1+x) + y^2(1+x)$$

$$\frac{dy}{dx} = (1+x) [1+y^2]$$

$$\int \frac{dy}{1+y^2} = \int (1+x) dx$$

$$\tan^{-1}(y) = x + \frac{x^2}{2} - \frac{3}{2}$$

$$\tan^{-1}y = x + \frac{x^2}{2} + C$$

$$y = \tan\left(x + \frac{x^2}{2} - \frac{3}{2}\right)$$

$$\tan^{-1}(0) = 1 + \frac{1}{2} + C$$

$$C = -\frac{3}{2} \quad \Rightarrow \quad \tan^{-1}y = x + \frac{x^2}{2} - \frac{3}{2}$$

Q No 3

$$(2xy + e^x) dx = e^x dy$$

$$\frac{y(2xy + e^x)}{e^x} = \frac{dy}{dx}$$

X

separable
is not
linear
is not

اور اب وہ linear ہے کہ linear ہے
is linear is linear

$$\frac{dy}{dx} + P(x)y = Q(x)$$

is not linear is not linear

is not linear is not linear

$$(2xy + e^x) dx = \frac{e^x}{y} dy$$

$$M dx + N dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial x} = \frac{e^x}{y}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$f(x) = \frac{M_1 - N_2}{N} = \frac{2x + e^x/y}{-e^x/y}$$

$$f(x) = \frac{2x + e^x/y}{-e^x/y} = \frac{2xy + e^x}{-e^x}$$

$$f(x) \neq \frac{2xy + e^x}{-e^x}$$

is not linear is not linear

$$f(y) = \frac{N_2 - M_1}{M} = \frac{e^x/y - 2x}{2xy + e^x}$$

$$= -\frac{(e^x + 2xy)}{e^x + 2xy}$$

$$= -\frac{1}{y}$$

$$g(y) = -\frac{1}{y}$$

$$-\int \frac{1}{y} dy = -\ln y = e^{-\ln y} = e^{\ln y^{-1}}$$

$$\mu = e$$

$$= y^{-1} = \frac{1}{y}$$

$$\mu = \frac{1}{y}$$

$y \cdot F$

$$\frac{1}{y} \int (2xy + e^x) dx - \frac{1}{y} \cdot \frac{e^x}{y} dy = 0$$

$$\int \frac{1}{y} (2xy + e^x) dx + \int 0 dy = c$$

$$\frac{1}{y} \left(2 \frac{x^2}{2} y + e^x \right) = c$$

$$x^2 + \frac{e^x}{y} = c$$

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QNo 3:-

$$y'' + 4y' + 3y = 0$$

find the value of m_1 & m_2

$$D^2 + 4D + 3 = 0$$

$$D^2 + D + 3D + 3 = 0$$

$$D(D+1) + 3(D+1) = 0$$

$$(D+3)(D+1) = 0$$

$$D = -1, -3$$

$$y(x) = C_1 e^{-3x} + C_2 e^{-x}$$

$$m_1 = -3$$

$$m_2 = -1$$

Derivative
($y' = \frac{dy}{dx}$) slopes

QNo 4:-

$$y = x - 1 + ce^{-x}$$

$$\frac{dy}{dx} = 1 + ce^{-x}(-1)$$

$$\frac{dy}{dx} = 1 - ce^{-x}$$

Orthogonal Trajectory

$$\frac{dy}{dx} = \frac{-1}{1 - ce^{-x}}$$

$$\int dy = - \int \frac{dx}{1 - ce^{-x}}$$

$$= - \int \frac{1}{1 - \frac{c}{e^x}} dx$$

$$= - \int \frac{e^x}{e^x - c} dx$$

$$= - \ln|e^x - c| + k$$

$$y = - \ln|e^x - c| + k$$

orthogonal
eq

$\Rightarrow$ Orthogonal
Trajectories

$\Rightarrow$ $\frac{dy}{dx} \perp \frac{dy}{dx}$ find

$\Rightarrow$ $\frac{dy}{dx} \perp \frac{dy}{dx}$ derivative

orthogonal Trajectory

$$\frac{dy}{dx} = \frac{-1}{1 - ce^{-x}}$$

value of derivative $\frac{dy}{dx}$ is $\frac{-1}{1 - ce^{-x}}$

orthogonal line perpendicular

$\Rightarrow$ $\frac{dy}{dx} \perp \frac{dy}{dx}$

m_1 and m_2 slopes

$m_2 \perp m_1$

$\Rightarrow$ derivative

$$m_1 m_2 = -1$$

$$\Rightarrow m_2 = \frac{-1}{m_1}$$

$$= \frac{e^{3x}}{25D^2} \left[x + \left(\frac{-D^2 x - 6Dx}{25} \right) \right]$$

$$= \frac{e^{3x}}{25D^2} \left[x + \left(\frac{0 - 6(1)}{25} \right) \right]$$

$$= \frac{e^{3x}}{25D^2} \left(x + \left(\frac{0 - 6(1)}{25} \right) \right)$$

$$= \frac{e^{3x}}{25D^2} \left(x - \frac{6}{25} \right)$$

$$= \frac{e^{3x}}{25D} \int \left(x - \frac{6}{25} \right) dx = \frac{e^{3x}}{25} \int \left(\frac{x^2}{2} - \frac{6}{25} x \right) dx$$

$$= \frac{e^{3x}}{25} \left[\frac{x^3}{3(2)} - \frac{6}{25} \cdot \frac{x^2}{2} \right]$$

$$= \frac{e^{3x}}{25} \left[\frac{x^3}{6} - \frac{3}{25} x^2 \right]$$

2nd term

$$= \frac{e^{3x}}{25D^2} \left(x + \frac{(-D^2x - 6Dx)}{25} \right)$$

$$= \frac{e^{3x}}{25D^2} \left[x + \frac{(0 - 6(1))}{25} \right]$$

$$= \frac{e^{3x}}{25D^2} \left(x + \frac{(0 - 6(1))}{25} \right)$$

$$= \frac{e^{3x}}{25D^2} \left(x - \frac{6}{25} \right)$$

$$= \frac{e^{3x}}{25D} \int \left(x - \frac{6}{25} \right) dx = \frac{e^{3x}}{25} \int \left(\frac{x^2}{2} - \frac{6}{25} x \right) dx$$

$$= \frac{e^{3x}}{25} \left[\frac{x^3}{3(2)} - \frac{6}{25} \cdot \frac{x^2}{2} \right]$$

$$= \frac{e^{3x}}{25} \left[\frac{x^3}{6} - \frac{3}{25} x^2 \right]$$

2nd term

Numerical
Analysis
ODE

$$\frac{e^{3x}}{25D^2} \left(1 + \frac{-D^2}{25} \right)$$

$$0.5 \leq x \leq 0.5$$

~~vi) viii)~~

$$y = \frac{x \sin 3x}{(D-3)^2 (D^2+16)}$$

$$= \frac{x \sin 3x}{D^4 + 16D^2 + 9D^2 + 144 - 6D^3 - 96D}$$

$$= \frac{x \sin 3x}{D^4 - 6D^3 + 25D^2 - 96D + 144}$$

using formula

$$= x \frac{\sin 3x}{D^4 - 6D^3 + 25D^2 - 96D + 144} - \frac{(4D^3 - 18D^2 + 50D - 96) \sin 3x}{(D^4 - 6D^3 + 25D^2 - 96D + 144)}$$

$$= x \frac{\sin 3x}{(-9)(-9) - 6(-9)D + 25(-9) - 96D + 144} - \frac{(4D^3 - 18D^2 + 50D - 96) \sin 3x}{(-9)(-9) - 6(-9)D + 25(-9) - 96D + 144}$$

$$= x \frac{\sin 3x}{-42D} - \frac{(4D^3 - 18D^2 + 50D - 96) \sin 3x}{(-42D)^2}$$

$$= -\frac{x}{42} \int \sin 3x dx - \frac{(4D^3 - 18D^2 + 50D - 96) \sin 3x}{(-42)(1764)(-9)}$$

$$= -\frac{x}{42} \left(\frac{-\cos 3x}{3} \right) + \frac{-108 \cos 3x + 182 \sin 3x + 150 \cos 3x}{15876} \quad \begin{array}{l} y = \sin 3x \\ y' = 3 \cos 3x \\ y'' = -9 \sin 3x \\ y''' = -27 \cos 3x \end{array}$$

$$= \frac{x \cos 3x}{126} + \frac{66 \sin 3x + 42 \cos 3x}{15876}$$

$$y_p = \frac{e^{3x}}{25} \left[\frac{x^2}{6} - \frac{6x^2}{50} \right] + \frac{x \cos 3x}{126} + \frac{66 \sin 3x + 42 \cos 3x}{15876}$$

$$y(x) = y_c(x) + y_p(x)$$

$$= (c_1 + c_2 x) e^x + c_3 \cos 4x + c_4 \sin 4x + \downarrow$$

[P.P = 2023]

Long

$$3x \cos \frac{y}{x} dy - \left(2x \sin \left(\frac{y}{x} \right) + 3y \cos \left(\frac{y}{x} \right) \right) dx = 0$$

÷ by x

Also by dx

$$3 \cos \left(\frac{y}{x} \right) \frac{dy}{\frac{y}{x}} - \left(2 \sin \left(\frac{y}{x} \right) + 3 \cos \left(\frac{y}{x} \right) \right) = 0$$

let $y = \frac{y}{x}$

$$y = xt$$

$$\frac{dy}{dx} = \left(x \frac{dt}{dx} + t \right)$$

$$3 \cos t \left(t + x \frac{dt}{dx} \right) = 2 \sin t + 3 t \cos t$$

$$3 t \cos t + 3 x \cos t \frac{dt}{dx} = 2 \sin t + 3 t \cos t$$

$$\frac{\cos t}{\sin t} dt = \frac{2}{3} \cdot \frac{1}{x} dx$$

derivative
y ke kisi
ke ln ke

$$\int \frac{\cos t}{\sin t} dt = \frac{2}{3} \int \frac{1}{x} dx$$

ab
y ke ln ke
ke ln ke

$$\ln \sin t = \frac{2}{3} \ln x + \ln c$$

$$\ln \sin t = \ln x^{2/3} + \ln c$$

$$\ln \sin t = \ln (x^{2/3} c)$$

$$\sin \left(\frac{y}{x} \right) = (x^{2/3} c)$$

method of variation of parameter

$$y'' + 4y = 4 \tan 2x \rightarrow Q$$

$$D^2 + 4 = 0$$

$$D = \pm 2i$$

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

ii

$$y_p(x) = u_1 y_1 + u_2 y_2$$

two parameter find u_1, u_2

$$y_1 = \cos 2x, \quad y_2 = \sin 2x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = \frac{2\cos^2 2x + 2\sin^2 2x}{2(1)} = 2 \neq 0$$

$$u_1 = - \int \frac{y_2 Q}{N} dx$$

$$= -2 \int \frac{\sin 2x \cdot 4 \sin 2x / \cos 2x}{2} dx$$

$$= -2 \int \frac{\sin^2 2x}{\cos 2x} dx = -2 \int \frac{1 - \cos^2 2x}{\cos 2x} dx$$

$$= -2 \int (\sec 2x - \cos 2x) dx = -2 \left(\ln |\sec 2x + \tan 2x| + \frac{\sin 2x}{2} \right)$$

$$u_1 = -\ln |\sec 2x + \tan 2x| + \sin 2x$$

$$u_2 = \int \frac{y_1 Q}{W} dx = \int \frac{\cos 2x \cdot 4 \sin 2x / \cos 2x}{2} dx$$

$$= -2 \int \frac{1}{\cos 2x} dx$$

$$u_2 = -\cos 2x$$

267 zero method
- 466x tail

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$$y_p(x) = u_1 y_1 + u_2 y_2$$

$$y_p(x) = \frac{(-\ln|\sec 2x + \tan 2x| + \sin 2x) / \cos 2x}{\cos 2x \sin 2x}$$

$$y_p(x) = -\cos 2x \ln|\sec 2x + \tan 2x|$$

Q :-

$$f(x) = 1$$

$$g(x) = \sin x$$

$$h(x) = \cos x$$

lin. Indep
or li Depen

$$W = \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 0 & -\sin x & -\cos x \end{vmatrix} \quad \text{Expanded by}$$

$$= 1 \begin{vmatrix} \cos x & -\sin x \\ -\sin x & -\cos x \end{vmatrix} = -\cos^2 x - \sin^2 x$$

$$= -(1) = -1 \neq 0$$

linearly Independent

P.P.
2023

method of variation
of parameters

$$Q_1 \quad \frac{dy}{dx} - 2\frac{dy}{dx} + y = e^x \sin^{-1} x$$

$$(D^2 - 2D + 1)y = e^x \sin^{-1} x$$

$$F(D)y = F(x)$$

For y_c put $F(D)y = 0$

Auxiliary equation of $f(m) = 0$

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m = 1, 1$$

$$\Rightarrow y_c = (C_1 + C_2 x)e^x \quad \text{--- (1)}$$

$$y_p = C_1 e^x + C_2 x e^x$$

$$y_p = u_1 e^x + u_2 x e^x \quad \text{--- (2)}$$

when

$$y_1 = e^x, \quad y_2 = x e^x$$

$$y_1' = e^x, \quad y_2' = e^x + x e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$= e^x (e^x + x e^x) - x e^x \cdot e^x$$

$$= e^{2x} + x e^{2x} - x e^{2x}$$

$$W = e^{2x}$$

$$u_1 = - \int \frac{y_2 F(x)}{W} dx$$

$$y_2 = x e^x, \quad W = e^{2x}, \quad F(x) = e^x \sin^{-1} x$$

$$u_1 = - \int x e^x \sin^{-1} x \, dx$$

$$u_1 = - \int x \sin^{-1} x \, dx$$

→ By integration parts

$$= - \left[\sin^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \frac{1}{\sqrt{1-x^2}} \, dx \right]$$

Inverse function
to function
1st function
Rakna hai

$$= - \left[\frac{x^2 \sin^{-1} x}{2} + \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} \, dx \right]$$

Formula

$$= - \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{-x^2}{\sqrt{1-x^2}} \, dx$$

→ 1st part
2nd part
1st part
2nd part
solve 1st

$$= - \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \left[\frac{(1-x^2) - 1}{\sqrt{1-x^2}} \right] \, dx$$

Subtraction
and adding '1'

$$= - \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \left(\frac{1-x^2}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \right) \, dx$$

$$= - \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \sqrt{1-x^2} \, dx + \frac{1}{2} \int \frac{1}{\sqrt{1-x^2}} \, dx$$

using formula

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x$$

$$\int \sqrt{a^2-x^2} \, dx = \frac{a^2 \sin^{-1} \frac{x}{a}}{2} + \frac{x}{2} \sqrt{a^2-x^2}$$

$$u_1 = - \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \left[\frac{1}{2} \sin^{-1} x + \frac{x}{2} \sqrt{1-x^2} \right] + \frac{1}{2} \sin^{-1} x$$

$$u_1 = -\frac{x^2}{2} \sin^{-1} x - \frac{1}{4} \sin^{-1} x - \frac{x}{4} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x$$

$$\Rightarrow u_1 = -\frac{x^2}{2} \sin^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{x}{4} \sqrt{1-x^2}$$

$$u_2 = \int \frac{y_1 + \ln x}{w} dx = \int \frac{e^x}{e^{2x}} e^x \sin^{-1} x dx$$

$$u_2 = \int \sin^{-1} x dx = \int \sin^{-1} x \cdot 1 dx$$

or integrating by parts

$$= (\sin^{-1} x) x - \int x \frac{1}{\sqrt{1-x^2}} dx$$

$$= x(\sin^{-1} x) + \frac{1}{2} \int (1-x^2)^{-1/2} (-2x) dx$$

$$= x \sin^{-1} x + \frac{1}{2} \frac{(1-x^2)^{-1/2+1}}{-1/2+1} = x \sin^{-1} x + \frac{(1-x^2)^{1/2}}{2} \cdot \frac{1}{1/2}$$

$$u_2 = x \sin^{-1} x + \sqrt{1-x^2}$$

$$y_p = u_1 e^x + u_2 x e^x$$

$$y_p = e^x \left[-\frac{x^2}{2} \sin^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{x}{4} \sqrt{1-x^2} \right] + x e^x \left[x \sin^{-1} x + \sqrt{1-x^2} \right]$$

$$y(x) = y_c + y_p$$

$$y(x) = (C_1 + C_2 x) e^x + e^x \left[-\frac{x^2}{2} \sin^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{x}{4} \sqrt{1-x^2} \right] + x e^x \left[x \sin^{-1} x + \sqrt{1-x^2} \right]$$